

EXPERIMENT 19: PLANETS DATABASE

AIM

To write a Prolog program to create a database of planets and retrieve information about planets using logical queries.

OBJECTIVE

- To understand the use of facts in Prolog.
- To create a simple planetary database.
- To store properties such as planet name, position and type.
- To retrieve planet information using queries.

ALGORITHM

1. Start.
2. Define facts for the planets.
3. Store the planet's position from the Sun.
4. Store the planet type.
5. Query the database for required information.
6. Display the matching result.
7. Stop.

SOURCE CODE

% Planets Database

planet(mercury, 1, terrestrial).

planet(venus, 2, terrestrial).

planet(earth, 3, terrestrial).

planet(mars, 4, terrestrial).

planet(jupiter, 5, gas_giant).

planet(saturn, 6, gas_giant).

planet(uranus, 7, ice_giant).

planet(neptune, 8, ice_giant).

Query

?- planet(earth, Position, Type).

```
?- [planets].  
% planets.pl compiled 0.00 sec, 8 clauses
```

```
?- planet(earth, Position, Type).
```

```
Position = 3,
```

```
Type = terrestrial.
```

```
true.
```

```
?- planet(Planet, 5, Type).
```

```
Planet = jupiter,
```

```
Type = gas_giant.
```

```
true.
```

```
?- ■
```

RESULT

Position = 3,

Type = terrestrial.

Another query:

?- planet(Planet, 5, Type).

Output:

Planet = jupiter,

Type = gas_giant.

The Prolog program successfully creates and queries a Planets Database.